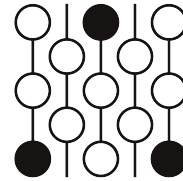


Review of Submission 9B

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

The combinatorial formula suggested for the coefficients $c_{(a,1^b)}(n)$ involves a pair (w, ρ) where w is a permutation and ρ is a partition.

The proof strategy starts from the result of Lentfer [3] giving the universality of the coefficients $c_\lambda(n)$, and that in the case of partitions λ of hook shapes, these coefficients can be extracted from the Hilbert series of the super coinvariant ring SR_n . It then uses the formula established by Murai–Rhoades–Wilson [4] for the graded Frobenius series for the algebra SR_n . An expression for the Hilbert series is then obtained by forgetting the S_n -module structure.

Here are some comments:

- The extra step of forgetting the S_n -module could be avoided by directly using the result of Rhoades–Wilson 2024 giving a formula for the Hilbert series.
- The equivalence between Eq. (2) and the formula established in [4] (see [4, Eqs. (1.13) and (2.15)]) which involves q -binomial coefficients, is not proved.
- Using the formula given in Eq. (2) rather than the formula provided in [4] seems unnecessary, since the purpose of Lemma 5 is to prove that Eq (2) can be expanded using q -binomial coefficients.

Correctness. The formula is correct, as is the proof.

Novelty. The solution uses the expected reference [3] to relate the coefficients in question to the ring SR_n . The fact of starting from the graded Frobenius series formula rather than the known formula for the Hilbert series makes the proof longer.

This extra step is also reflected in the combinatorial interpretation which involves a permutation w and a partition ρ that are not genuinely related, and which makes the formula seem slightly less “natural”.

2 Attribution

Lemma 4 is a classical result. I suggest citing standard textbooks references rather than reproving it.

3 Recommendation

I recommend publication after minor revisions (see in particular the comment after Eq. (2)).

4 Submission with inline comments

Overview / Strategy

The proof passes from the ordinary diagonal coinvariant algebra with m bosonic alphabets to the one-bosonic-one-fermionic coinvariant algebra SR_n . A diagonal supersymmetry theorem says that the same universal Schur coefficients $c_\lambda(n)$ occur after replacing $s_\lambda(q_1, \dots, q_m)$ by the super Schur function $s_\lambda(q/u)$. In one bosonic and one fermionic variable, every nonempty non-hook shape vanishes, while

$$s_{(a,1^b)}(q/u) = q^a u^b + q^{a-1} u^{b+1}.$$

This gives a triangular relation between the hook generating functions $C_b(q) = \sum_{a \geq 1} c_{(a,1^b)}(n) q^a$ and the coefficients $D_b(q) = [u^b] \text{Hilb}(SR_n; q, u)$.

The second input is the Fields Frobenius formula, proved by Murai–Rhoades–Wilson. After forgetting the S_n -module structure and using the Robinson–Schensted descent correspondence, this formula becomes a sum over permutations $w \in S_n$. Extracting the coefficient of u^b introduces a Gaussian binomial coefficient. The triangular relation is then solved by induction using the q -Pascal identity, and the Gaussian binomial is finally expanded as a generating function for partitions in a rectangle. This yields the stated counting formula for hook coefficients.

4.1 Definitions and headline theorem

For $m, n \geq 1$, let

$$P_n^{(m)} = \mathbb{C}[x_{i,j} : 1 \leq i \leq n, 1 \leq j \leq m],$$

with the diagonal action of S_n given by $\sigma \cdot x_{i,j} = x_{\sigma(i),j}$. The coinvariant algebra with m sets of n variables is

$$R_n^{(m)} = P_n^{(m)} / \left\langle \left(P_n^{(m)} \right)_+^{S_n} \right\rangle,$$

where $\left(P_n^{(m)} \right)_+^{S_n}$ denotes the invariant polynomials with zero constant term. The algebra is m -multigraded by degrees in the m sets of variables, and

$$\text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) = \sum_{i_1, \dots, i_m \geq 0} \dim(R_n^{(m)})_{(i_1, \dots, i_m)} q_1^{i_1} \cdots q_m^{i_m}.$$

As in the problem statement, write

$$\text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) = \sum_{\lambda} c_\lambda(n) s_\lambda(q_1, \dots, q_m)$$

for the known universal Schur expansion, with coefficients $c_\lambda(n)$ independent of m .

For a permutation $w = w_1 \cdots w_n \in S_n$, define

$$\text{Des}(w) = \{i \in \{1, \dots, n-1\} : w_i > w_{i+1}\}, \quad \text{des}(w) = |\text{Des}(w)|, \quad \text{maj}(w) = \sum_{i \in \text{Des}(w)} i.$$

For integers $b, r \geq 0$, the notation $\rho \subseteq b \times r$ means that ρ is a partition with at most b parts, each at most r . Its size is $|\rho|$.

Theorem 1 (Original problem: hook coefficients in the universal Hilbert series). *Fix $n \geq 1$. Then*

$$c_{\emptyset}(n) = 1.$$

For every nonempty hook $\lambda = (a, 1^b)$, where $a \geq 1$ and $b \geq 0$,

$$c_{(a, 1^b)}(n) = \# \left\{ (w, \rho) : \begin{array}{l} w \in S_n, \quad d = \text{des}(w) \geq b+1, \\ \rho \subseteq b \times (d-b-1), \\ a = \text{maj}(w) - bd + \binom{b}{2} + |\rho| \end{array} \right\}.$$

When $b = 0$, the rectangle $0 \times (d-1)$ contains only the empty partition, and the formula becomes

$$c_{(a)}(n) = \#\{w \in S_n : \text{des}(w) \geq 1, \text{maj}(w) = a\} \quad (a \geq 1),$$

with the identity permutation accounted for separately by $c_{\emptyset}(n) = 1$. Also,

$$c_{(a, 1^b)}(n) = 0 \quad \text{whenever } b \geq n-1.$$

The proof of Theorem 1 occupies the rest of the document.

4.2 Cited inputs

We use two external theorems.

First, define the one-bosonic-one-fermionic coinvariant algebra

$$SR_n = \frac{\mathbb{C}[x_1, \dots, x_n] \otimes \Lambda(\theta_1, \dots, \theta_n)}{\langle (\mathbb{C}[x_1, \dots, x_n] \otimes \Lambda(\theta_1, \dots, \theta_n))_+^{S_n} \rangle},$$

where S_n acts diagonally by permuting the x_i 's and the θ_i 's. The variable q records x -degree and u records θ -degree. The diagonal supersymmetry theorem for bosonic-fermionic coinvariant rings [3] implies that the same universal coefficients $c_{\lambda}(n)$ satisfy

$$\text{Hilb}(SR_n; q, u) = \sum_{\lambda} c_{\lambda}(n) s_{\lambda}(q/u), \quad (1)$$

where $s_{\lambda}(q/u)$ is the one-bosonic-one-fermionic super Schur function.

Second, the Fields Frobenius formula, proved by Murai–Rhoades–Wilson [4], gives the bigraded Frobenius characteristic of SR_n . For a standard Young tableau T , let

$$\text{Des}(T) = \{i : i+1 \text{ appears in a strictly lower row of } T \text{ than } i\},$$

and define $\text{des}(T) = |\text{Des}(T)|$, $\text{maj}(T) = \sum_{i \in \text{Des}(T)} i$. Then

$$\text{grFrob}(SR_n; q, u) = \sum_{T \in \text{SYT}(n)} q^{\text{maj}(T) - \binom{\text{des}(T)+1}{2}} \prod_{i=1}^{\text{des}(T)} (q^i + u) s_{\text{shape}(T)}. \quad (2)$$

This citation is not precise: the formula does not appear in [4] in this form.

Here $\text{SYT}(n)$ denotes the set of all standard Young tableaux with n boxes, and $s_{\text{shape}(T)}$ is the Schur function appearing in the Frobenius characteristic. This is the Schur-tableau form of the $t = 0$ Delta expression; see also [2]. We also use the standard Robinson–Schensted correspondence and its descent property $\text{Des}(Q(w)) = \text{Des}(w)$ for the recording tableau $Q(w)$; see Fulton [1].

4.3 Forgetting the S_n -module structure

Lemma 2. *The Hilbert series of SR_n is*

$$\text{Hilb}(SR_n; q, u) = \sum_{w \in S_n} q^{\text{maj}(w) - \binom{\text{des}(w)+1}{2}} \prod_{i=1}^{\text{des}(w)} (q^i + u).$$

Proof. Let $f^\mu = |\text{SYT}(\mu)|$. The Frobenius characteristic sends the irreducible S_n -module of shape μ to s_μ . Therefore applying the dimension map

$$s_\mu \longmapsto f^\mu$$

to a graded Frobenius characteristic gives the corresponding Hilbert series. Applying this map to (2) gives

$$\text{Hilb}(SR_n; q, u) = \sum_{\mu \vdash n} \sum_{T \in \text{SYT}(\mu)} f^\mu q^{\text{maj}(T) - \binom{\text{des}(T)+1}{2}} \prod_{i=1}^{\text{des}(T)} (q^i + u).$$

For each tableau $T \in \text{SYT}(\mu)$, the factor f^μ counts a choice of another tableau $P \in \text{SYT}(\mu)$. Thus the right-hand side is a sum over ordered pairs (P, Q) of standard Young tableaux of the same shape, with $Q = T$, weighted by

$$q^{\text{maj}(Q) - \binom{\text{des}(Q)+1}{2}} \prod_{i=1}^{\text{des}(Q)} (q^i + u).$$

By the Robinson–Schensted correspondence, such pairs (P, Q) are in bijection with permutations $w \in S_n$. Under this bijection, $\text{Des}(Q(w)) = \text{Des}(w)$, hence $\text{des}(Q(w)) = \text{des}(w)$ and $\text{maj}(Q(w)) = \text{maj}(w)$. Substituting these equalities into the preceding sum gives

$$\text{Hilb}(SR_n; q, u) = \sum_{w \in S_n} q^{\text{maj}(w) - \binom{\text{des}(w)+1}{2}} \prod_{i=1}^{\text{des}(w)} (q^i + u),$$

as claimed. □

4.4 The one-bosonic-one-fermionic specialization

Lemma 3. *With the convention*

$$s_\lambda(q/u) = \sum_{\nu \subseteq \lambda} s_\nu(q) s_{\lambda'/\nu'}(u),$$

one has

$$s_{\emptyset}(q/u) = 1.$$

If $\lambda = (a, 1^b)$ is a hook with $a \geq 1, b \geq 0$, then

$$s_{(a, 1^b)}(q/u) = q^a u^b + q^{a-1} u^{b+1}.$$

Every nonempty non-hook shape has $s_{\lambda}(q/u) = 0$.

Proof. For an ordinary Schur function in one variable, $s_{\nu}(q)$ is nonzero only when ν has at most one row. Indeed, in the semistandard tableau definition, only the entry 1 is available, and a column of length at least two would violate strict increase down columns. Thus $s_{(r)}(q) = q^r$, with $(0) = \emptyset$.

Similarly, a skew Schur function $s_{\alpha/\beta}(u)$ in one variable is nonzero exactly when the skew shape α/β has at most one cell in each column, i.e. is a horizontal strip. In that case the unique semistandard filling uses only 1's and contributes $u^{|\alpha/\beta|}$.

For $\lambda = \emptyset$, the only summand is $\nu = \emptyset$, giving $s_{\emptyset}(q/u) = 1$.

Now suppose $\lambda \neq \emptyset$. If λ is not a hook, then $\lambda_2 \geq 2$. In any potentially nonzero summand, ν must be a row, say $\nu = (r)$, so $\nu' = (1^r)$ is contained in the first column of λ' . Removing ν' from λ' does not affect the second column of λ' , which has length $\lambda_2 \geq 2$. Therefore λ'/ν' has at least two cells in one column and is not a horizontal strip. Hence every summand is zero.

It remains to compute the hook case. Let $\lambda = (a, 1^b)$, so $\lambda' = (b+1, 1^{a-1})$. Again, ν must be a row (r) with $0 \leq r \leq a$. Removing $\nu' = (1^r)$ from the first column of λ' , the remaining part of that first column has length $a-r$. The skew shape λ'/ν' is a horizontal strip if and only if $a-r \leq 1$, i.e. $r = a$ or $r = a-1$. For $r = a$, the contribution is $q^a u^b$. For $r = a-1$, the contribution is $q^{a-1} u^{b+1}$. These are the only nonzero summands, proving the formula. $\square$

4.5 Gaussian binomial identities

For integers r, s , define

$$\begin{bmatrix} r \\ s \end{bmatrix}_q = \begin{cases} \sum_{\rho \subseteq s \times (r-s)} q^{|\rho|}, & 0 \leq s \leq r, \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 4. *The following identities hold.*

1. If $0 \leq b \leq d$, then

$$[u^b] \prod_{i=1}^d (q^i + u) = q^{\binom{d-b+1}{2}} \begin{bmatrix} d \\ b \end{bmatrix}_q.$$

If $b > d$, both sides are 0, using the convention above.

2. For $1 \leq b \leq d$,

$$\begin{bmatrix} d \\ b \end{bmatrix}_q = \begin{bmatrix} d-1 \\ b \end{bmatrix}_q + q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q.$$

Proof. For the first identity, if $b > d$, no term in the product can contribute u^b , and $\begin{bmatrix} d \\ b \end{bmatrix}_q = 0$. Assume $0 \leq b \leq d$, and set $r = d - b$. To obtain the coefficient of u^b , choose r indices

$$1 \leq t_1 < \cdots < t_r \leq d$$

from which the q^i -term is selected. The corresponding contribution is $q^{t_1 + \cdots + t_r}$. Define

$$\rho_j = t_{r+1-j} - (r + 1 - j) \quad (1 \leq j \leq r).$$

Then $\rho_1 \geq \cdots \geq \rho_r \geq 0$, and $\rho_1 \leq d - r = b$, so $\rho \subseteq r \times b$. Conversely, every partition in $r \times b$ arises uniquely in this way. Moreover,

$$t_1 + \cdots + t_r = 1 + \cdots + r + |\rho| = \binom{r+1}{2} + |\rho|.$$

Thus

$$[u^b] \prod_{i=1}^d (q^i + u) = q^{\binom{r+1}{2}} \sum_{\rho \subseteq r \times b} q^{|\rho|}.$$

Conjugation of partitions is a size-preserving bijection between partitions in $r \times b$ and partitions in $b \times r$. Hence

$$\sum_{\rho \subseteq r \times b} q^{|\rho|} = \begin{bmatrix} d \\ b \end{bmatrix}_q,$$

because $r = d - b$. This proves the first identity.

For the second identity, if $b = d$, it reads $1 = 0 + 1$, which is immediate. Assume $1 \leq b < d$. Partitions in the rectangle $b \times (d - b)$ split into two disjoint classes. If the first part is at most $d - b - 1$, the partition lies in $b \times (d - b - 1)$, and these partitions contribute $\begin{bmatrix} d-1 \\ b \end{bmatrix}_q$. If the first part is exactly $d - b$, removing that first part leaves a partition in $(b - 1) \times (d - b)$, and the removed part contributes a factor q^{d-b} . These partitions therefore contribute $q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q$. The two cases are disjoint and exhaustive, proving the q -Pascal identity. $\square$

4.6 The coefficients of u^b

For $b \geq 0$, define

$$D_b(q) = [u^b] \text{Hilb}(SR_n; q, u).$$

Lemma 5. For every $b \geq 0$,

$$D_b(q) = \sum_{w \in S_n} q^{\text{maj}(w) - b \text{des}(w) + \binom{b}{2}} \begin{bmatrix} \text{des}(w) \\ b \end{bmatrix}_q.$$

Proof. Use Lemma 2. Fix $w \in S_n$, and write

$$d = \text{des}(w), \quad M = \text{maj}(w).$$

The contribution of w to $\text{Hilb}(SR_n; q, u)$ is

$$q^{M - \binom{d+1}{2}} \prod_{i=1}^d (q^i + u).$$

By Lemma 4,

$$[u^b] \prod_{i=1}^d (q^i + u) = q^{\binom{d-b+1}{2}} \begin{bmatrix} d \\ b \end{bmatrix}_q,$$

with the understanding that the expression is zero when $b > d$. Multiplying by the prefactor gives

$$q^{M - \binom{d+1}{2}} q^{\binom{d-b+1}{2}} \begin{bmatrix} d \\ b \end{bmatrix}_q.$$

Since

$$-\binom{d+1}{2} + \binom{d-b+1}{2} = -bd + \binom{b}{2},$$

this contribution equals

$$q^{M - bd + \binom{b}{2}} \begin{bmatrix} d \\ b \end{bmatrix}_q.$$

Summing over all $w \in S_n$ proves the formula. □

4.7 Solving the triangular relation

For $b \geq 0$, define the hook generating polynomial

$$C_b(q) = \sum_{a \geq 1} c_{(a, 1^b)}(n) q^a,$$

and set $C_{-1}(q) = 0$.

Proposition 6. *One has $c_{\emptyset}(n) = 1$. Moreover, for every $b \geq 0$,*

$$C_b(q) = \sum_{\substack{w \in S_n \\ \text{des}(w) \geq b+1}} q^{\text{maj}(w) - b \text{des}(w) + \binom{b}{2}} \begin{bmatrix} \text{des}(w) - 1 \\ b \end{bmatrix}_q.$$

Proof. First, the ideal defining $R_n^{(m)}$ is homogeneous and generated by positive-degree elements, so the degree-zero component of $R_n^{(m)}$ is $\mathbb{C}$. Hence

$$\text{Hilb}(R_n^{(m)}; 0, \dots, 0) = 1.$$

In the universal Schur expansion, $s_{\emptyset} = 1$, while every nonempty Schur polynomial has zero constant term. Therefore $c_{\emptyset}(n) = 1$.

By (1) and Lemma 3,

$$\text{Hilb}(SR_n; q, u) = 1 + \sum_{b \geq 0} \sum_{a \geq 1} c_{(a, 1^b)}(n) (q^a u^b + q^{a-1} u^{b+1}),$$

because all nonempty non-hook shapes vanish under the specialization q/u . Taking the coefficient of u^b gives

$$D_b(q) = \delta_{b,0} + C_b(q) + q^{-1} C_{b-1}(q), \quad (3)$$

where $\delta_{b,0}$ is the Kronecker delta and $C_{-1}(q) = 0$.

We now prove the displayed formula for $C_b(q)$ by induction on b .

For $b = 0$, equation (3) gives

$$C_0(q) = D_0(q) - 1.$$

By Lemma 5,

$$D_0(q) = \sum_{w \in S_n} q^{\text{maj}(w)}.$$

The unique permutation with no descents is $12 \cdots n$, and it contributes 1. Every other permutation has at least one descent. Therefore

$$C_0(q) = \sum_{\substack{w \in S_n \\ \text{des}(w) \geq 1}} q^{\text{maj}(w)}.$$

This is the required formula for $b = 0$, since $\left[\begin{smallmatrix} \text{des}(w)-1 \\ 0 \end{smallmatrix} \right]_q = 1$ whenever $\text{des}(w) \geq 1$.

Now let $b \geq 1$, and assume the formula is known for $b - 1$. From (3),

$$C_b(q) = D_b(q) - q^{-1} C_{b-1}(q).$$

Fix $w \in S_n$, and write $d = \text{des}(w)$, $M = \text{maj}(w)$. By Lemma 5, the contribution of w to $D_b(q)$ is

$$q^{M-bd+\binom{b}{2}} \left[\begin{smallmatrix} d \\ b \end{smallmatrix} \right]_q.$$

By the induction hypothesis, the contribution of w to $q^{-1} C_{b-1}(q)$, when $d \geq b$, is

$$q^{-1} q^{M-(b-1)d+\binom{b-1}{2}} \left[\begin{smallmatrix} d-1 \\ b-1 \end{smallmatrix} \right]_q.$$

The exponent satisfies

$$-1 + M - (b-1)d + \binom{b-1}{2} = M - bd + \binom{b}{2} + d - b.$$

Thus, for all w with $d \geq b$, the difference of the two contributions is

$$q^{M-bd+\binom{b}{2}} \left(\left[\begin{smallmatrix} d \\ b \end{smallmatrix} \right]_q - q^{d-b} \left[\begin{smallmatrix} d-1 \\ b-1 \end{smallmatrix} \right]_q \right).$$

If $d < b$, both contributions are zero. For $d \geq b$, Lemma 4 gives

$$\begin{bmatrix} d \\ b \end{bmatrix}_q - q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q = \begin{bmatrix} d-1 \\ b \end{bmatrix}_q.$$

Therefore

$$C_b(q) = \sum_{w \in S_n} q^{M-bd + \binom{b}{2}} \begin{bmatrix} d-1 \\ b \end{bmatrix}_q.$$

The Gaussian binomial $\begin{bmatrix} d-1 \\ b \end{bmatrix}_q$ is zero unless $d-1 \geq b$, i.e. unless $d \geq b+1$. Hence

$$C_b(q) = \sum_{\substack{w \in S_n \\ \text{des}(w) \geq b+1}} q^{\text{maj}(w) - b \text{des}(w) + \binom{b}{2}} \begin{bmatrix} \text{des}(w) - 1 \\ b \end{bmatrix}_q,$$

which completes the induction. □

4.8 Proof of the headline theorem

Proof of Theorem 1. The equality $c_{\emptyset}(n) = 1$ is part of Proposition 6.

Fix $b \geq 0$. Proposition 6 gives

$$C_b(q) = \sum_{\substack{w \in S_n \\ d = \text{des}(w) \geq b+1}} q^{\text{maj}(w) - bd + \binom{b}{2}} \begin{bmatrix} d-1 \\ b \end{bmatrix}_q.$$

By the definition of the Gaussian binomial coefficient,

$$\begin{bmatrix} d-1 \\ b \end{bmatrix}_q = \sum_{\rho \subseteq b \times (d-b-1)} q^{|\rho|},$$

because $(d-1) - b = d - b - 1$.

The last line can be omitted.

Reviewer 1

Therefore

$$C_b(q) = \sum_{\substack{w \in S_n \\ d = \text{des}(w) \geq b+1}} \sum_{\rho \subseteq b \times (d-b-1)} q^{\text{maj}(w) - bd + \binom{b}{2} + |\rho|}.$$

On the other hand,

$$C_b(q) = \sum_{a \geq 1} c_{(a, 1^b)}(n) q^a.$$

Thus the coefficient of q^a in $C_b(q)$ is exactly the number of pairs (w, ρ) such that

$$w \in S_n, \quad d = \text{des}(w) \geq b+1,$$

$$\rho \subseteq b \times (d - b - 1),$$

and

$$a = \text{maj}(w) - bd + \binom{b}{2} + |\rho|.$$

This is precisely the boxed formula in the theorem.

It remains only to verify the stated edge cases. First, the exponent just displayed is always positive under the condition $d \geq b + 1$. Indeed, if the descent set of w has size d , say

$$\text{Des}(w) = \{i_1 < \cdots < i_d\},$$

then $i_j \geq j$ for every j , so

$$\text{maj}(w) = i_1 + \cdots + i_d \geq 1 + \cdots + d = \binom{d+1}{2}.$$

Consequently,

$$\text{maj}(w) - bd + \binom{b}{2} \geq \binom{d+1}{2} - bd + \binom{b}{2} = \binom{d-b+1}{2}.$$

Since $d \geq b + 1$, the last binomial coefficient is at least 1, and adding $|\rho| \geq 0$ keeps the exponent positive.

When $b = 0$, the rectangle $0 \times (d - 1)$ contains only the empty partition. The condition $d \geq b + 1$ becomes $\text{des}(w) \geq 1$, and the equation for a becomes $a = \text{maj}(w)$. Hence

$$c_{(a)}(n) = \#\{w \in S_n : \text{des}(w) \geq 1, \text{maj}(w) = a\} \quad (a \geq 1).$$

The identity permutation has $\text{des} = 0$ and $\text{maj} = 0$, and its contribution is the separate term $c_{\emptyset}(n) = 1$.

Finally, every permutation in S_n has at most $n - 1$ descents. If $b \geq n - 1$, then the required inequality $d = \text{des}(w) \geq b + 1$ is impossible. Therefore

$$c_{(a,1^b)}(n) = 0 \quad \text{for all } b \geq n - 1.$$

The theorem is proved. □

References

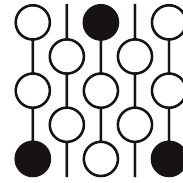
- [1] W. Fulton, *Young Tableaux: With Applications to Representation Theory and Geometry*, London Mathematical Society Student Texts, vol. 35, Cambridge University Press, 1997.
- [2] J. Haglund, B. Rhoades, and M. Shimozono, *Hall–Littlewood expansions of Schur delta operators at $t = 0$* , Séminaire Lotharingien de Combinatoire **79** (2019), Article B79c.
- [3] J. Lentfer, *Diagonal Supersymmetry for Coinvariant Rings*, arXiv:2505.14885, 2025. <https://arxiv.org/abs/2505.14885>
- [4] S. Murai, B. Rhoades, and A. Wilson, *A proof of the Fields Conjectures*, arXiv:2505.24027, 2025. <https://arxiv.org/abs/2505.24027>

Review of Submission 9B

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The submission appears to be entirely correct, as far as I can tell.

Summary. This submission proves the coefficients $\{c_\lambda(n) : \lambda \text{ is a hook}\}$ count pairs (w, ρ) of permutations w and partitions ρ satisfying several conditions depending on λ , n , and the descents of the permutation w . The main steps of the proof are:

1. Using diagonal supersymmetry, show that the Hilbert series for the *superspace*¹ coinvariant ring can be written as sums of various $\{C_b\}$ where $C_b = \sum_{a \geq 1} c_{(a, 1^b)}(n)q^a$.
2. Uses the Murai–Rhoades–Wilson Frobenius series for the superspace coinvariant ring to derive another formula for its Hilbert series.
3. By comparing the formulations for the Hilbert series in (1) and (2), and using induction, it obtains another formula for C_b , involving permutations and descents.
4. Prove by q -identities that that the proposed combinatorial interpretation matches the coefficient of q^a in the formulation of C_b in (3).

Attributions. Unlike the others, this submission does *not* seem to plagiarize! On the other hand, it does seem to be unaware of key papers in the field (such as the Rhoades–Wilson Forum of Math. Pi. '24 paper). The knowledge of this paper would likely have simplified some of its arguments (like in §3.3). It also includes proofs of a couple of classical or routine lemmas (like q -Pascal identities), which it could instead cite references for.

Novelty. Although correct, the “combinatorial interpretation” in some ways feels a bit contrived. The objects which are claimed to be counted (pairs of a permutation and partition satisfying certain conditions) only arise as indexes in the sums after manipulating q -series, and are not really “used” in the proof at all.

On the other hand, this interpretation is quite different from the other four I’ve seen: the three other model solutions and the human solution. The other four all involve ordered set partitions,

¹one-commuting, one-anticommuting variable

whereas this one does not – likely because it seemed unaware of the Rhoades–Wilson paper. This opens the (potentially easy, but maybe interesting) question for finding a bijection between the objects in this combinatorial interpretation and any of the others.

Exposition. The writing style is clear and easy-to-follow. It could easily be shortened if necessary (it reproves some classical formulas and overexplains some points), but I still found it the easiest to read.

2 Recommendation

I would categorize this solutions as publishable with minor revisions.

3 Submission with inline comments

Overview / Strategy

The proof passes from the ordinary diagonal coinvariant algebra with m bosonic alphabets to the one-bosonic-one-fermionic coinvariant algebra SR_n . A diagonal supersymmetry theorem says that the same universal Schur coefficients $c_\lambda(n)$ occur after replacing $s_\lambda(q_1, \dots, q_m)$ by the super Schur function $s_\lambda(q/u)$. In one bosonic and one fermionic variable, every nonempty non-hook shape vanishes, while

$$s_{(a,1^b)}(q/u) = q^a u^b + q^{a-1} u^{b+1}.$$

This gives a triangular relation between the hook generating functions $C_b(q) = \sum_{a \geq 1} c_{(a,1^b)}(n) q^a$ and the coefficients $D_b(q) = [u^b] \text{Hilb}(SR_n; q, u)$.

The second input is the Fields Frobenius formula, proved by Murai–Rhoades–Wilson. After forgetting the S_n -module structure and using the Robinson–Schensted descent correspondence, this formula becomes a sum over permutations $w \in S_n$. Extracting the coefficient of u^b introduces a Gaussian binomial coefficient. The triangular relation is then solved by induction using the q -Pascal identity, and the Gaussian binomial is finally expanded as a generating function for partitions in a rectangle. This yields the stated counting formula for hook coefficients.

3.1 Definitions and headline theorem

For $m, n \geq 1$, let

$$P_n^{(m)} = \mathbb{C}[x_{i,j} : 1 \leq i \leq n, 1 \leq j \leq m],$$

with the diagonal action of S_n given by $\sigma \cdot x_{i,j} = x_{\sigma(i),j}$. The coinvariant algebra with m sets of n variables is

$$R_n^{(m)} = P_n^{(m)} / \left\langle \left(P_n^{(m)} \right)_+^{S_n} \right\rangle,$$

where $(P_n^{(m)})_+^{S_n}$ denotes the invariant polynomials with zero constant term. The algebra is m -multigraded by degrees in the m sets of variables, and

$$\text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) = \sum_{i_1, \dots, i_m \geq 0} \dim(R_n^{(m)})_{(i_1, \dots, i_m)} q_1^{i_1} \cdots q_m^{i_m}.$$

As in the problem statement, write

$$\text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) = \sum_{\lambda} c_{\lambda}(n) s_{\lambda}(q_1, \dots, q_m)$$

for the known universal Schur expansion, with coefficients $c_{\lambda}(n)$ independent of m .

For a permutation $w = w_1 \cdots w_n \in S_n$, define

$$\text{Des}(w) = \{i \in \{1, \dots, n-1\} : w_i > w_{i+1}\}, \quad \text{des}(w) = |\text{Des}(w)|, \quad \text{maj}(w) = \sum_{i \in \text{Des}(w)} i.$$

For integers $b, r \geq 0$, the notation $\rho \subseteq b \times r$ means that ρ is a partition with at most b parts, each at most r . Its size is $|\rho|$.

Theorem 1 (Original problem: hook coefficients in the universal Hilbert series). *Fix $n \geq 1$. Then*

$$c_{\emptyset}(n) = 1.$$

For every nonempty hook $\lambda = (a, 1^b)$, where $a \geq 1$ and $b \geq 0$,

$$c_{(a, 1^b)}(n) = \# \left\{ (w, \rho) : \begin{array}{l} w \in S_n, \quad d = \text{des}(w) \geq b+1, \\ \rho \subseteq b \times (d-b-1), \\ a = \text{maj}(w) - bd + \binom{b}{2} + |\rho| \end{array} \right\}.$$

When $b = 0$, the rectangle $0 \times (d-1)$ contains only the empty partition, and the formula becomes

$$c_{(a)}(n) = \#\{w \in S_n : \text{des}(w) \geq 1, \text{maj}(w) = a\} \quad (a \geq 1),$$

with the identity permutation accounted for separately by $c_{\emptyset}(n) = 1$. Also,

$$c_{(a, 1^b)}(n) = 0 \quad \text{whenever } b \geq n-1.$$

The proof of Theorem 1 occupies the rest of the document.

3.2 Cited inputs

We use two external theorems.

First, define the one-bosonic-one-fermionic coinvariant algebra

$$SR_n = \frac{\mathbb{C}[x_1, \dots, x_n] \otimes \Lambda(\theta_1, \dots, \theta_n)}{\langle (\mathbb{C}[x_1, \dots, x_n] \otimes \Lambda(\theta_1, \dots, \theta_n))_+^{S_n} \rangle},$$

where S_n acts diagonally by permuting the x_i 's and the θ_i 's. The variable q records x -degree and u records θ -degree. The diagonal supersymmetry theorem for bosonic-fermionic coinvariant rings [3] implies that the same universal coefficients $c_\lambda(n)$ satisfy

$$\text{Hilb}(SR_n; q, u) = \sum_{\lambda} c_{\lambda}(n) s_{\lambda}(q/u), \quad (1)$$

where $s_{\lambda}(q/u)$ is the one-bosonic-one-fermionic super Schur function.

Second, the Fields Frobenius formula, proved by Murai–Rhoades–Wilson [4], gives the bigraded Frobenius characteristic of SR_n . For a standard Young tableau T , let

$$\text{Des}(T) = \{ i : i + 1 \text{ appears in a strictly lower row of } T \text{ than } i \},$$

and define $\text{des}(T) = |\text{Des}(T)|$, $\text{maj}(T) = \sum_{i \in \text{Des}(T)} i$. Then

$$\text{grFrob}(SR_n; q, u) = \sum_{T \in \text{SYT}(n)} q^{\text{maj}(T) - \binom{\text{des}(T)+1}{2}} \prod_{i=1}^{\text{des}(T)} (q^i + u) s_{\text{shape}(T)}. \quad (2)$$

Reviewer 2

This formula is a little different than the one appearing in Murai–Rhoades–Wilson. It seems to actually be equivalent, but this is not immediate and should be explained.

Here $\text{SYT}(n)$ denotes the set of all standard Young tableaux with n boxes, and $s_{\text{shape}(T)}$ is the Schur function appearing in the Frobenius characteristic. This is the Schur-tableau form of the $t = 0$ Delta expression; see also [2].

Reviewer 2

An unnecessary detail (and the Delta expression is undefined), but this does not really matter.

We also use the standard Robinson–Schensted correspondence and its descent property $\text{Des}(Q(w)) = \text{Des}(w)$ for the recording tableau $Q(w)$; see Fulton [1].

3.3 Forgetting the S_n -module structure

Reviewer 2

Seems unnecessary to derive the Hilbert series structure by starting with the Frobenius series, since the Hilbert series is already explicitly proved by Rhoades–Wilson. However, this way does give a different formula for the Hilbert series, which is interesting, as it leads to a different combinatorial interpretation. Is it clear how the two formulas for the Hilbert series are equivalent?

Lemma 2. *The Hilbert series of SR_n is*

$$\text{Hilb}(SR_n; q, u) = \sum_{w \in S_n} q^{\text{maj}(w) - \binom{\text{des}(w)+1}{2}} \prod_{i=1}^{\text{des}(w)} (q^i + u).$$

Proof. Let $f^\mu = |\text{SYT}(\mu)|$. The Frobenius characteristic sends the irreducible S_n -module of shape μ to s_μ . Therefore applying the dimension map

$$s_\mu \longmapsto f^\mu$$

to a graded Frobenius characteristic gives the corresponding Hilbert series. Applying this map to (2) gives

$$\text{Hilb}(SR_n; q, u) = \sum_{\mu \vdash n} \sum_{T \in \text{SYT}(\mu)} f^\mu q^{\text{maj}(T) - \binom{\text{des}(T)+1}{2}} \prod_{i=1}^{\text{des}(T)} (q^i + u).$$

For each tableau $T \in \text{SYT}(\mu)$, the factor f^μ counts a choice of another tableau $P \in \text{SYT}(\mu)$. Thus the right-hand side is a sum over ordered pairs (P, Q) of standard Young tableaux of the same shape, with $Q = T$, weighted by

$$q^{\text{maj}(Q) - \binom{\text{des}(Q)+1}{2}} \prod_{i=1}^{\text{des}(Q)} (q^i + u).$$

By the Robinson–Schensted correspondence, such pairs (P, Q) are in bijection with permutations $w \in S_n$. Under this bijection, $\text{Des}(Q(w)) = \text{Des}(w)$, hence $\text{des}(Q(w)) = \text{des}(w)$ and $\text{maj}(Q(w)) = \text{maj}(w)$. Substituting these equalities into the preceding sum gives

$$\text{Hilb}(SR_n; q, u) = \sum_{w \in S_n} q^{\text{maj}(w) - \binom{\text{des}(w)+1}{2}} \prod_{i=1}^{\text{des}(w)} (q^i + u),$$

as claimed. □

3.4 The one-bosonic-one-fermionic specialization

Reviewer 2

Maybe don't need proof of this fact, pretty straightforward for the reader to check, and likely classical.

Lemma 3. *With the convention*

$$s_\lambda(q/u) = \sum_{\nu \subseteq \lambda} s_\nu(q) s_{\lambda'/\nu'}(u),$$

one has

$$s_\emptyset(q/u) = 1.$$

If $\lambda = (a, 1^b)$ is a hook with $a \geq 1$, $b \geq 0$, then

$$s_{(a, 1^b)}(q/u) = q^a u^b + q^{a-1} u^{b+1}.$$

Every nonempty non-hook shape has $s_\lambda(q/u) = 0$.

Proof. For an ordinary Schur function in one variable, $s_\nu(q)$ is nonzero only when ν has at most one row. Indeed, in the semistandard tableau definition, only the entry 1 is available, and a column of length at least two would violate strict increase down columns. Thus $s_{(r)}(q) = q^r$, with $(0) = \emptyset$.

Similarly, a skew Schur function $s_{\alpha/\beta}(u)$ in one variable is nonzero exactly when the skew shape α/β has at most one cell in each column, i.e. is a horizontal strip. In that case the unique semistandard filling uses only 1's and contributes $u^{|\alpha/\beta|}$.

For $\lambda = \emptyset$, the only summand is $\nu = \emptyset$, giving $s_\emptyset(q/u) = 1$.

Now suppose $\lambda \neq \emptyset$. If λ is not a hook, then $\lambda_2 \geq 2$. In any potentially nonzero summand, ν must be a row, say $\nu = (r)$, so $\nu' = (1^r)$ is contained in the first column of λ' . Removing ν' from λ' does not affect the second column of λ' , which has length $\lambda_2 \geq 2$. Therefore λ'/ν' has at least two cells in one column and is not a horizontal strip. Hence every summand is zero.

It remains to compute the hook case. Let $\lambda = (a, 1^b)$, so $\lambda' = (b+1, 1^{a-1})$. Again, ν must be a row (r) with $0 \leq r \leq a$. Removing $\nu' = (1^r)$ from the first column of λ' , the remaining part of that first column has length $a - r$. The skew shape λ'/ν' is a horizontal strip if and only if $a - r \leq 1$, i.e. $r = a$ or $r = a - 1$. For $r = a$, the contribution is $q^a u^b$. For $r = a - 1$, the contribution is $q^{a-1} u^{b+1}$. These are the only nonzero summands, proving the formula. $\square$

3.5 Gaussian binomial identities

For integers r, s , define

$$\begin{bmatrix} r \\ s \end{bmatrix}_q = \begin{cases} \sum_{\rho \subseteq s \times (r-s)} q^{|\rho|}, & 0 \leq s \leq r, \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 4. *The following identities hold.*

Reviewer 2

Also seem well-known, especially part 2

1. If $0 \leq b \leq d$, then

$$[u^b] \prod_{i=1}^d (q^i + u) = q^{\binom{d-b+1}{2}} \begin{bmatrix} d \\ b \end{bmatrix}_q.$$

If $b > d$, both sides are 0, using the convention above.

2. For $1 \leq b \leq d$,

$$\begin{bmatrix} d \\ b \end{bmatrix}_q = \begin{bmatrix} d-1 \\ b \end{bmatrix}_q + q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q.$$

Proof. For the first identity, if $b > d$, no term in the product can contribute u^b , and $\begin{bmatrix} d \\ b \end{bmatrix}_q = 0$. Assume $0 \leq b \leq d$, and set $r = d - b$. To obtain the coefficient of u^b , choose r indices

$$1 \leq t_1 < \cdots < t_r \leq d$$

from which the q^i -term is selected. The corresponding contribution is $q^{t_1+\dots+t_r}$. Define

$$\rho_j = t_{r+1-j} - (r + 1 - j) \quad (1 \leq j \leq r).$$

Then $\rho_1 \geq \dots \geq \rho_r \geq 0$, and $\rho_1 \leq d - r = b$, so $\rho \subseteq r \times b$. Conversely, every partition in $r \times b$ arises uniquely in this way. Moreover,

$$t_1 + \dots + t_r = 1 + \dots + r + |\rho| = \binom{r+1}{2} + |\rho|.$$

Thus

$$[u^b] \prod_{i=1}^d (q^i + u) = q^{\binom{r+1}{2}} \sum_{\rho \subseteq r \times b} q^{|\rho|}.$$

Conjugation of partitions is a size-preserving bijection between partitions in $r \times b$ and partitions in $b \times r$. Hence

$$\sum_{\rho \subseteq r \times b} q^{|\rho|} = \begin{bmatrix} d \\ b \end{bmatrix}_q,$$

because $r = d - b$. This proves the first identity.

For the second identity, if $b = d$, it reads $1 = 0 + 1$, which is immediate. Assume $1 \leq b < d$. Partitions in the rectangle $b \times (d - b)$ split into two disjoint classes. If the first part is at most $d - b - 1$, the partition lies in $b \times (d - b - 1)$, and these partitions contribute $\begin{bmatrix} d-1 \\ b \end{bmatrix}_q$. If the first part is exactly $d - b$, removing that first part leaves a partition in $(b - 1) \times (d - b)$, and the removed part contributes a factor q^{d-b} . These partitions therefore contribute $q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q$. The two cases are disjoint and exhaustive, proving the q -Pascal identity. $\square$

3.6 The coefficients of u^b

For $b \geq 0$, define

$$D_b(q) = [u^b] \text{Hilb}(SR_n; q, u).$$

Lemma 5. *For every $b \geq 0$,*

$$D_b(q) = \sum_{w \in S_n} q^{\text{maj}(w) - b \text{des}(w) + \binom{b}{2}} \begin{bmatrix} \text{des}(w) \\ b \end{bmatrix}_q.$$

Proof. Use Lemma 2. Fix $w \in S_n$, and write

Reviewer 2

Extremely minor / stylistic note, but seems unnecessary to shorten des and maj to d and M , especially given that you switch the notation back (and then back again!) later.

$$d = \text{des}(w), \quad M = \text{maj}(w).$$

The contribution of w to $\text{Hilb}(SR_n; q, u)$ is

$$q^{M - \binom{d+1}{2}} \prod_{i=1}^d (q^i + u).$$

By Lemma 4,

$$[u^b] \prod_{i=1}^d (q^i + u) = q^{\binom{d-b+1}{2}} \left[\begin{matrix} d \\ b \end{matrix} \right]_q,$$

with the understanding that the expression is zero when $b > d$. Multiplying by the prefactor gives

$$q^{M - \binom{d+1}{2}} q^{\binom{d-b+1}{2}} \left[\begin{matrix} d \\ b \end{matrix} \right]_q.$$

Since

$$-\binom{d+1}{2} + \binom{d-b+1}{2} = -bd + \binom{b}{2},$$

this contribution equals

$$q^{M - bd + \binom{b}{2}} \left[\begin{matrix} d \\ b \end{matrix} \right]_q.$$

Summing over all $w \in S_n$ proves the formula. □

3.7 Solving the triangular relation

For $b \geq 0$, define the hook generating polynomial

$$C_b(q) = \sum_{a \geq 1} c_{(a, 1^b)}(n) q^a,$$

and set $C_{-1}(q) = 0$.

Proposition 6. *One has $c_{\emptyset}(n) = 1$. Moreover, for every $b \geq 0$,*

$$C_b(q) = \sum_{\substack{w \in S_n \\ \text{des}(w) \geq b+1}} q^{\text{maj}(w) - b \text{des}(w) + \binom{b}{2}} \left[\begin{matrix} \text{des}(w) - 1 \\ b \end{matrix} \right]_q.$$

Proof. First, the ideal defining $R_n^{(m)}$ is homogeneous and generated by positive-degree elements, so the degree-zero component of $R_n^{(m)}$ is $\mathbb{C}$. Hence

$$\text{Hilb}(R_n^{(m)}; 0, \dots, 0) = 1.$$

In the universal Schur expansion, $s_{\emptyset} = 1$, while every nonempty Schur polynomial has zero constant term. Therefore $c_{\emptyset}(n) = 1$.

By (1) and Lemma 3,

$$\text{Hilb}(SR_n; q, u) = 1 + \sum_{b \geq 0} \sum_{a \geq 1} c_{(a, 1^b)}(n) (q^a u^b + q^{a-1} u^{b+1}),$$

because all nonempty non-hook shapes vanish under the specialization q/u . Taking the coefficient of u^b gives

$$D_b(q) = \delta_{b,0} + C_b(q) + q^{-1} C_{b-1}(q), \quad (3)$$

where $\delta_{b,0}$ is the Kronecker delta and $C_{-1}(q) = 0$.

We now prove the displayed formula for $C_b(q)$ by induction on b .

For $b = 0$, equation (3) gives

$$C_0(q) = D_0(q) - 1.$$

By Lemma 5,

$$D_0(q) = \sum_{w \in S_n} q^{\text{maj}(w)}.$$

The unique permutation with no descents is $12 \cdots n$, and it contributes 1. Every other permutation has at least one descent. Therefore

$$C_0(q) = \sum_{\substack{w \in S_n \\ \text{des}(w) \geq 1}} q^{\text{maj}(w)}.$$

This is the required formula for $b = 0$, since $\left[\begin{smallmatrix} \text{des}(w)-1 \\ 0 \end{smallmatrix} \right]_q = 1$ whenever $\text{des}(w) \geq 1$.

Now let $b \geq 1$, and assume the formula is known for $b - 1$. From (3),

$$C_b(q) = D_b(q) - q^{-1} C_{b-1}(q).$$

Fix $w \in S_n$, and write $d = \text{des}(w)$, $M = \text{maj}(w)$. By Lemma 5, the contribution of w to $D_b(q)$ is

$$q^{M-bd+\binom{b}{2}} \left[\begin{smallmatrix} d \\ b \end{smallmatrix} \right]_q.$$

By the induction hypothesis, the contribution of w to $q^{-1} C_{b-1}(q)$, when $d \geq b$, is

$$q^{-1} q^{M-(b-1)d+\binom{b-1}{2}} \left[\begin{smallmatrix} d-1 \\ b-1 \end{smallmatrix} \right]_q.$$

The exponent satisfies

$$-1 + M - (b-1)d + \binom{b-1}{2} = M - bd + \binom{b}{2} + d - b.$$

Thus, for all w with $d \geq b$, the difference of the two contributions is

$$q^{M-bd+\binom{b}{2}} \left(\left[\begin{smallmatrix} d \\ b \end{smallmatrix} \right]_q - q^{d-b} \left[\begin{smallmatrix} d-1 \\ b-1 \end{smallmatrix} \right]_q \right).$$

It's straightforward to check this is correct, but interesting stylistically that this doesn't have any intermediate steps here, but other much simpler things are explained in other places, like that $(d-1)-b = d-b-1$.

If $d < b$, both contributions are zero. For $d \geq b$, Lemma 4 gives

$$\begin{bmatrix} d \\ b \end{bmatrix}_q - q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q = \begin{bmatrix} d-1 \\ b \end{bmatrix}_q.$$

Therefore

$$C_b(q) = \sum_{w \in S_n} q^{\text{maj}(w) - bd + \binom{b}{2}} \begin{bmatrix} d-1 \\ b \end{bmatrix}_q.$$

The Gaussian binomial $\begin{bmatrix} d-1 \\ b \end{bmatrix}_q$ is zero unless $d-1 \geq b$, i.e. unless $d \geq b+1$. Hence

$$C_b(q) = \sum_{\substack{w \in S_n \\ \text{des}(w) \geq b+1}} q^{\text{maj}(w) - b \text{des}(w) + \binom{b}{2}} \begin{bmatrix} \text{des}(w) - 1 \\ b \end{bmatrix}_q,$$

which completes the induction. □

3.8 Proof of the headline theorem

Proof of Theorem 1. The equality $c_{\emptyset}(n) = 1$ is part of Proposition 6.

Fix $b \geq 0$. Proposition 6 gives

$$C_b(q) = \sum_{\substack{w \in S_n \\ d = \text{des}(w) \geq b+1}} q^{\text{maj}(w) - bd + \binom{b}{2}} \begin{bmatrix} d-1 \\ b \end{bmatrix}_q.$$

By the definition of the Gaussian binomial coefficient,

$$\begin{bmatrix} d-1 \\ b \end{bmatrix}_q = \sum_{\rho \subseteq b \times (d-b-1)} q^{|\rho|},$$

because $(d-1)-b = d-b-1$. Therefore

$$C_b(q) = \sum_{\substack{w \in S_n \\ d = \text{des}(w) \geq b+1}} \sum_{\rho \subseteq b \times (d-b-1)} q^{\text{maj}(w) - bd + \binom{b}{2} + |\rho|}.$$

On the other hand,

$$C_b(q) = \sum_{a \geq 1} c_{(a, 1^b)}(n) q^a.$$

Thus the coefficient of q^a in $C_b(q)$ is exactly the number of pairs (w, ρ) such that

$$w \in S_n, \quad d = \text{des}(w) \geq b+1,$$

$$\rho \subseteq b \times (d - b - 1),$$

and

$$a = \text{maj}(w) - bd + \binom{b}{2} + |\rho|.$$

This is precisely the boxed formula in the theorem.

It remains only to verify the stated edge cases. First, the exponent just displayed is always positive under the condition $d \geq b + 1$. Indeed, if the descent set of w has size d , say

$$\text{Des}(w) = \{i_1 < \cdots < i_d\},$$

then $i_j \geq j$ for every j , so

$$\text{maj}(w) = i_1 + \cdots + i_d \geq 1 + \cdots + d = \binom{d+1}{2}.$$

Consequently,

$$\text{maj}(w) - bd + \binom{b}{2} \geq \binom{d+1}{2} - bd + \binom{b}{2} = \binom{d-b+1}{2}.$$

Since $d \geq b + 1$, the last binomial coefficient is at least 1, and adding $|\rho| \geq 0$ keeps the exponent positive.

When $b = 0$, the rectangle $0 \times (d - 1)$ contains only the empty partition. The condition $d \geq b + 1$ becomes $\text{des}(w) \geq 1$, and the equation for a becomes $a = \text{maj}(w)$. Hence

$$c_{(a)}(n) = \#\{w \in S_n : \text{des}(w) \geq 1, \text{maj}(w) = a\} \quad (a \geq 1).$$

The identity permutation has $\text{des} = 0$ and $\text{maj} = 0$, and its contribution is the separate term $c_{\emptyset}(n) = 1$.

Finally, every permutation in S_n has at most $n - 1$ descents. If $b \geq n - 1$, then the required inequality $d = \text{des}(w) \geq b + 1$ is impossible. Therefore

$$c_{(a,1^b)}(n) = 0 \quad \text{for all } b \geq n - 1.$$

The theorem is proved. □

References

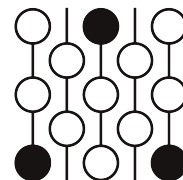
- [1] W. Fulton, *Young Tableaux: With Applications to Representation Theory and Geometry*, London Mathematical Society Student Texts, vol. 35, Cambridge University Press, 1997.
- [2] J. Haglund, B. Rhoades, and M. Shimozono, *Hall–Littlewood expansions of Schur delta operators at $t = 0$* , Séminaire Lotharingien de Combinatoire **79** (2019), Article B79c.
- [3] J. Lentfer, *Diagonal Supersymmetry for Coinvariant Rings*, arXiv:2505.14885, 2025. <https://arxiv.org/abs/2505.14885>
- [4] S. Murai, B. Rhoades, and A. Wilson, *A proof of the Fields Conjectures*, arXiv:2505.24027, 2025. <https://arxiv.org/abs/2505.24027>

Review of Submission 9B

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

Correctness. The mathematical content in this paper is correct, though occasionally needlessly roundabout.

Novelty. This paper is novel in that it comes up with a combinatorial interpretation that, notably, does not use ordered set partitions, likely because it seems unaware of Rhoades–Wilson (2024) or Sagan–Swanson (2024).

That said, the interpretation mixes two combinatorial objects, permutations and partitions, in an unnatural way, simply because the partition appears in a q -binomial expansion. Plus, the several added conditions are cumbersome. I doubt that this object would appear elsewhere in the literature or be useful elsewhere in the field.

Attribution. The citations provided in this paper are correct. Some classical results, though, should be better acknowledged (and do not need to be reproven). There is one equation that is inexplicably altered from its initial source but is nevertheless equivalent and therefore still correct.

2 Recommendation

This paper is publishable with minor revisions. The unnecessary detour into the Frobenius characteristic and the citation of this result should be fixed. The classical results should be acknowledged and not reproven.

3 Submission with inline comments

Overview / Strategy

The proof passes from the ordinary diagonal coinvariant algebra with m bosonic alphabets to the one-bosonic-one-fermionic coinvariant algebra SR_n . A diagonal supersymmetry theorem says that the same universal Schur coefficients $c_\lambda(n)$ occur after replacing $s_\lambda(q_1, \dots, q_m)$ by the super Schur

function $s_\lambda(q/u)$. In one bosonic and one fermionic variable, every nonempty non-hook shape vanishes, while

$$s_{(a,1^b)}(q/u) = q^a u^b + q^{a-1} u^{b+1}.$$

This gives a triangular relation between the hook generating functions $C_b(q) = \sum_{a \geq 1} c_{(a,1^b)}(n) q^a$ and the coefficients $D_b(q) = [u^b] \text{Hilb}(SR_n; q, u)$.

The second input is the Fields Frobenius formula, proved by Murai–Rhoades–Wilson. After forgetting the S_n -module structure and using the Robinson–Schensted descent correspondence, this formula becomes a sum over permutations $w \in S_n$. Extracting the coefficient of u^b introduces a Gaussian binomial coefficient. The triangular relation is then solved by induction using the q -Pascal identity, and the Gaussian binomial is finally expanded as a generating function for partitions in a rectangle. This yields the stated counting formula for hook coefficients.

3.1 Definitions and headline theorem

Reviewer 3

Stylistically, one wouldn't use the term "headline theorem."

For $m, n \geq 1$, let

$$P_n^{(m)} = \mathbb{C}[x_{i,j} : 1 \leq i \leq n, 1 \leq j \leq m],$$

with the diagonal action of S_n given by $\sigma \cdot x_{i,j} = x_{\sigma(i),j}$. The coinvariant algebra with m sets of n variables is

$$R_n^{(m)} = P_n^{(m)} / \left\langle \left(P_n^{(m)} \right)_+^{S_n} \right\rangle,$$

where $\left(P_n^{(m)} \right)_+^{S_n}$ denotes the invariant polynomials with zero constant term. The algebra is m -multigraded by degrees in the m sets of variables, and

$$\text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) = \sum_{i_1, \dots, i_m \geq 0} \dim \left(R_n^{(m)} \right)_{(i_1, \dots, i_m)} q_1^{i_1} \cdots q_m^{i_m}.$$

As in the problem statement, write

$$\text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) = \sum_{\lambda} c_{\lambda}(n) s_{\lambda}(q_1, \dots, q_m)$$

for the known universal Schur expansion, with coefficients $c_{\lambda}(n)$ independent of m .

For a permutation $w = w_1 \cdots w_n \in S_n$, define

$$\text{Des}(w) = \{ i \in \{1, \dots, n-1\} : w_i > w_{i+1} \}, \quad \text{des}(w) = |\text{Des}(w)|, \quad \text{maj}(w) = \sum_{i \in \text{Des}(w)} i.$$

For integers $b, r \geq 0$, the notation $\rho \subseteq b \times r$ means that ρ is a partition with at most b parts, each at most r . Its size is $|\rho|$.

Theorem 1 (Original problem: hook coefficients in the universal Hilbert series). *Fix $n \geq 1$. Then*

$$c_{\emptyset}(n) = 1.$$

For every nonempty hook $\lambda = (a, 1^b)$, where $a \geq 1$ and $b \geq 0$,

$$c_{(a, 1^b)}(n) = \# \left\{ (w, \rho) : \begin{array}{l} w \in S_n, \quad d = \text{des}(w) \geq b + 1, \\ \rho \subseteq b \times (d - b - 1), \\ a = \text{maj}(w) - bd + \binom{b}{2} + |\rho| \end{array} \right\}.$$

When $b = 0$, the rectangle $0 \times (d - 1)$ contains only the empty partition, and the formula becomes

$$c_{(a)}(n) = \# \{ w \in S_n : \text{des}(w) \geq 1, \text{maj}(w) = a \} \quad (a \geq 1),$$

with the identity permutation accounted for separately by $c_{\emptyset}(n) = 1$. Also,

$$c_{(a, 1^b)}(n) = 0 \quad \text{whenever } b \geq n - 1.$$

The proof of Theorem 1 occupies the rest of the document.

3.2 Cited inputs

We use two external theorems.

First, define the one-bosonic-one-fermionic coinvariant algebra

$$SR_n = \frac{\mathbb{C}[x_1, \dots, x_n] \otimes \Lambda(\theta_1, \dots, \theta_n)}{\langle (\mathbb{C}[x_1, \dots, x_n] \otimes \Lambda(\theta_1, \dots, \theta_n))_+^{S_n} \rangle},$$

where S_n acts diagonally by permuting the x_i 's and the θ_i 's. The variable q records x -degree and u records θ -degree. The diagonal supersymmetry theorem for bosonic-fermionic coinvariant rings [3] implies that the same universal coefficients $c_{\lambda}(n)$ satisfy

$$\text{Hilb}(SR_n; q, u) = \sum_{\lambda} c_{\lambda}(n) s_{\lambda}(q/u), \quad (1)$$

where $s_{\lambda}(q/u)$ is the one-bosonic-one-fermionic super Schur function.

Second, the Fields Frobenius formula, proved by Murai–Rhoades–Wilson [4], gives the bigraded Frobenius characteristic of SR_n . For a standard Young tableau T , let

$$\text{Des}(T) = \{ i : i + 1 \text{ appears in a strictly lower row of } T \text{ than } i \},$$

and define $\text{des}(T) = |\text{Des}(T)|$, $\text{maj}(T) = \sum_{i \in \text{Des}(T)} i$. Then

$$\text{grFrob}(SR_n; q, u) = \sum_{T \in \text{SYT}(n)} q^{\text{maj}(T) - \binom{\text{des}(T)+1}{2}} \prod_{i=1}^{\text{des}(T)} (q^i + u) s_{\text{shape}(T)}. \quad (2)$$

Interestingly, the above equation does not appear in the cited paper, but instead a different equation that can be shown to be the same via a routine calculation. Then we pass from the Frobenius characteristic to the Hilbert series by forgetting the module structure, when it would be both simpler and easier to just start with the Hilbert result, as stated in Rhoades–Wilson. My guess is that this paper is unaware of that result, which also explains why it does not end up at an ordered-set-partition interpretation.

Here $\text{SYT}(n)$ denotes the set of all standard Young tableaux with n boxes, and $s_{\text{shape}(T)}$ is the Schur function appearing in the Frobenius characteristic. This is the Schur-tableau form of the $t = 0$ Delta expression; see also [2]. We also use the standard Robinson–Schensted correspondence and its descent property $\text{Des}(Q(w)) = \text{Des}(w)$ for the recording tableau $Q(w)$; see Fulton [1].

3.3 Forgetting the S_n -module structure

Lemma 2. *The Hilbert series of SR_n is*

$$\text{Hilb}(SR_n; q, u) = \sum_{w \in S_n} q^{\text{maj}(w) - \binom{\text{des}(w)+1}{2}} \prod_{i=1}^{\text{des}(w)} (q^i + u).$$

Proof. Let $f^\mu = |\text{SYT}(\mu)|$. The Frobenius characteristic sends the irreducible S_n -module of shape μ to s_μ . Therefore applying the dimension map

$$s_\mu \longmapsto f^\mu$$

to a graded Frobenius characteristic gives the corresponding Hilbert series. Applying this map to (2) gives

$$\text{Hilb}(SR_n; q, u) = \sum_{\mu \vdash n} \sum_{T \in \text{SYT}(\mu)} f^\mu q^{\text{maj}(T) - \binom{\text{des}(T)+1}{2}} \prod_{i=1}^{\text{des}(T)} (q^i + u).$$

For each tableau $T \in \text{SYT}(\mu)$, the factor f^μ counts a choice of another tableau $P \in \text{SYT}(\mu)$. Thus the right-hand side is a sum over ordered pairs (P, Q) of standard Young tableaux of the same shape, with $Q = T$, weighted by

$$q^{\text{maj}(Q) - \binom{\text{des}(Q)+1}{2}} \prod_{i=1}^{\text{des}(Q)} (q^i + u).$$

By the Robinson–Schensted correspondence, such pairs (P, Q) are in bijection with permutations $w \in S_n$. Under this bijection, $\text{Des}(Q(w)) = \text{Des}(w)$, hence $\text{des}(Q(w)) = \text{des}(w)$ and $\text{maj}(Q(w)) = \text{maj}(w)$. Substituting these equalities into the preceding sum gives

$$\text{Hilb}(SR_n; q, u) = \sum_{w \in S_n} q^{\text{maj}(w) - \binom{\text{des}(w)+1}{2}} \prod_{i=1}^{\text{des}(w)} (q^i + u),$$

as claimed. □

3.4 The one-bosonic-one-fermionic specialization

Lemma 3. *With the convention*

$$s_\lambda(q/u) = \sum_{\nu \subseteq \lambda} s_\nu(q) s_{\lambda'/\nu'}(u),$$

one has

$$s_\emptyset(q/u) = 1.$$

If $\lambda = (a, 1^b)$ is a hook with $a \geq 1$, $b \geq 0$, then

$$s_{(a, 1^b)}(q/u) = q^a u^b + q^{a-1} u^{b+1}.$$

Every nonempty non-hook shape has $s_\lambda(q/u) = 0$.

Proof. For an ordinary Schur function in one variable, $s_\nu(q)$ is nonzero only when ν has at most one row. Indeed, in the semistandard tableau definition, only the entry 1 is available, and a column of length at least two would violate strict increase down columns. Thus $s_{(r)}(q) = q^r$, with $(0) = \emptyset$.

Similarly, a skew Schur function $s_{\alpha/\beta}(u)$ in one variable is nonzero exactly when the skew shape α/β has at most one cell in each column, i.e. is a horizontal strip. In that case the unique semistandard filling uses only 1's and contributes $u^{|\alpha/\beta|}$.

For $\lambda = \emptyset$, the only summand is $\nu = \emptyset$, giving $s_\emptyset(q/u) = 1$.

Now suppose $\lambda \neq \emptyset$. If λ is not a hook, then $\lambda_2 \geq 2$. In any potentially nonzero summand, ν must be a row, say $\nu = (r)$, so $\nu' = (1^r)$ is contained in the first column of λ' . Removing ν' from λ' does not affect the second column of λ' , which has length $\lambda_2 \geq 2$. Therefore λ'/ν' has at least two cells in one column and is not a horizontal strip. Hence every summand is zero.

It remains to compute the hook case. Let $\lambda = (a, 1^b)$, so $\lambda' = (b+1, 1^{a-1})$. Again, ν must be a row (r) with $0 \leq r \leq a$. Removing $\nu' = (1^r)$ from the first column of λ' , the remaining part of that first column has length $a - r$. The skew shape λ'/ν' is a horizontal strip if and only if $a - r \leq 1$, i.e. $r = a$ or $r = a - 1$. For $r = a$, the contribution is $q^a u^b$. For $r = a - 1$, the contribution is $q^{a-1} u^{b+1}$. These are the only nonzero summands, proving the formula. $\square$

Reviewer 3

I believe this result is classical, which should be stated, or if there is a source, cited.

3.5 Gaussian binomial identities

For integers r, s , define

$$\begin{bmatrix} r \\ s \end{bmatrix}_q = \begin{cases} \sum_{\rho \subseteq s \times (r-s)} q^{|\rho|}, & 0 \leq s \leq r, \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 4. *The following identities hold.*

1. If $0 \leq b \leq d$, then

$$[u^b] \prod_{i=1}^d (q^i + u) = q^{\binom{d-b+1}{2}} \begin{bmatrix} d \\ b \end{bmatrix}_q.$$

If $b > d$, both sides are 0, using the convention above.

2. For $1 \leq b \leq d$,

$$\begin{bmatrix} d \\ b \end{bmatrix}_q = \begin{bmatrix} d-1 \\ b \end{bmatrix}_q + q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q.$$

Proof. For the first identity, if $b > d$, no term in the product can contribute u^b , and $\begin{bmatrix} d \\ b \end{bmatrix}_q = 0$. Assume $0 \leq b \leq d$, and set $r = d - b$. To obtain the coefficient of u^b , choose r indices

$$1 \leq t_1 < \cdots < t_r \leq d$$

from which the q^i -term is selected. The corresponding contribution is $q^{t_1 + \cdots + t_r}$. Define

$$\rho_j = t_{r+1-j} - (r + 1 - j) \quad (1 \leq j \leq r).$$

Then $\rho_1 \geq \cdots \geq \rho_r \geq 0$, and $\rho_1 \leq d - r = b$, so $\rho \subseteq r \times b$. Conversely, every partition in $r \times b$ arises uniquely in this way. Moreover,

$$t_1 + \cdots + t_r = 1 + \cdots + r + |\rho| = \binom{r+1}{2} + |\rho|.$$

Thus

$$[u^b] \prod_{i=1}^d (q^i + u) = q^{\binom{r+1}{2}} \sum_{\rho \subseteq r \times b} q^{|\rho|}.$$

Conjugation of partitions is a size-preserving bijection between partitions in $r \times b$ and partitions in $b \times r$. Hence

$$\sum_{\rho \subseteq r \times b} q^{|\rho|} = \begin{bmatrix} d \\ b \end{bmatrix}_q,$$

because $r = d - b$. This proves the first identity.

For the second identity, if $b = d$, it reads $1 = 0 + 1$, which is immediate. Assume $1 \leq b < d$. Partitions in the rectangle $b \times (d - b)$ split into two disjoint classes. If the first part is at most $d - b - 1$, the partition lies in $b \times (d - b - 1)$, and these partitions contribute $\begin{bmatrix} d-1 \\ b \end{bmatrix}_q$. If the first part is exactly $d - b$, removing that first part leaves a partition in $(b - 1) \times (d - b)$, and the removed part contributes a factor q^{d-b} . These partitions therefore contribute $q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q$. The two cases are disjoint and exhaustive, proving the q -Pascal identity. $\square$

Reviewer 3

The first identity follows directly from the q -Binomial theorem, and the second (as the author alludes to) is the q -Pascal identity. As both of these are well-known, they should be properly cited as such and do

not need to be reproven here.

3.6 The coefficients of u^b

For $b \geq 0$, define

$$D_b(q) = [u^b] \text{Hilb}(SR_n; q, u).$$

Lemma 5. *For every $b \geq 0$,*

$$D_b(q) = \sum_{w \in S_n} q^{\text{maj}(w) - b \text{des}(w) + \binom{b}{2}} \left[\begin{matrix} \text{des}(w) \\ b \end{matrix} \right]_q.$$

Proof. Use Lemma 2. Fix $w \in S_n$, and write

$$d = \text{des}(w), \quad M = \text{maj}(w).$$

The contribution of w to $\text{Hilb}(SR_n; q, u)$ is

$$q^{M - \binom{d+1}{2}} \prod_{i=1}^d (q^i + u).$$

By Lemma 4,

$$[u^b] \prod_{i=1}^d (q^i + u) = q^{\binom{d-b+1}{2}} \left[\begin{matrix} d \\ b \end{matrix} \right]_q,$$

with the understanding that the expression is zero when $b > d$. Multiplying by the prefactor gives

$$q^{M - \binom{d+1}{2}} q^{\binom{d-b+1}{2}} \left[\begin{matrix} d \\ b \end{matrix} \right]_q.$$

Since

$$-\binom{d+1}{2} + \binom{d-b+1}{2} = -bd + \binom{b}{2},$$

this contribution equals

$$q^{M - bd + \binom{b}{2}} \left[\begin{matrix} d \\ b \end{matrix} \right]_q.$$

Summing over all $w \in S_n$ proves the formula. □

3.7 Solving the triangular relation

For $b \geq 0$, define the hook generating polynomial

$$C_b(q) = \sum_{a \geq 1} c_{(a,1^b)}(n) q^a,$$

and set $C_{-1}(q) = 0$.

Proposition 6. *One has $c_{\emptyset}(n) = 1$. Moreover, for every $b \geq 0$,*

$$C_b(q) = \sum_{\substack{w \in S_n \\ \text{des}(w) \geq b+1}} q^{\text{maj}(w) - b \text{des}(w) + \binom{b}{2}} \left[\begin{matrix} \text{des}(w) - 1 \\ b \end{matrix} \right]_q.$$

Proof. First, the ideal defining $R_n^{(m)}$ is homogeneous and generated by positive-degree elements, so the degree-zero component of $R_n^{(m)}$ is $\mathbb{C}$. Hence

$$\text{Hilb}(R_n^{(m)}; 0, \dots, 0) = 1.$$

In the universal Schur expansion, $s_{\emptyset} = 1$, while every nonempty Schur polynomial has zero constant term. Therefore $c_{\emptyset}(n) = 1$.

By (1) and Lemma 3,

$$\text{Hilb}(SR_n; q, u) = 1 + \sum_{b \geq 0} \sum_{a \geq 1} c_{(a,1^b)}(n) (q^a u^b + q^{a-1} u^{b+1}),$$

because all nonempty non-hook shapes vanish under the specialization q/u . Taking the coefficient of u^b gives

$$D_b(q) = \delta_{b,0} + C_b(q) + q^{-1} C_{b-1}(q), \quad (3)$$

where $\delta_{b,0}$ is the Kronecker delta and $C_{-1}(q) = 0$.

We now prove the displayed formula for $C_b(q)$ by induction on b .

For $b = 0$, equation (3) gives

$$C_0(q) = D_0(q) - 1.$$

By Lemma 5,

$$D_0(q) = \sum_{w \in S_n} q^{\text{maj}(w)}.$$

The unique permutation with no descents is $12 \cdots n$, and it contributes 1. Every other permutation has at least one descent. Therefore

$$C_0(q) = \sum_{\substack{w \in S_n \\ \text{des}(w) \geq 1}} q^{\text{maj}(w)}.$$

This is the required formula for $b = 0$, since $\left[\begin{matrix} \text{des}(w) - 1 \\ 0 \end{matrix} \right]_q = 1$ whenever $\text{des}(w) \geq 1$.

Now let $b \geq 1$, and assume the formula is known for $b - 1$. From (3),

$$C_b(q) = D_b(q) - q^{-1}C_{b-1}(q).$$

Fix $w \in S_n$, and write $d = \text{des}(w)$, $M = \text{maj}(w)$. By Lemma 5, the contribution of w to $D_b(q)$ is

$$q^{M-bd+\binom{b}{2}} \begin{bmatrix} d \\ b \end{bmatrix}_q.$$

By the induction hypothesis, the contribution of w to $q^{-1}C_{b-1}(q)$, when $d \geq b$, is

$$q^{-1}q^{M-(b-1)d+\binom{b-1}{2}} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q.$$

The exponent satisfies

$$-1 + M - (b-1)d + \binom{b-1}{2} = M - bd + \binom{b}{2} + d - b.$$

Thus, for all w with $d \geq b$, the difference of the two contributions is

$$q^{M-bd+\binom{b}{2}} \left(\begin{bmatrix} d \\ b \end{bmatrix}_q - q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q \right).$$

If $d < b$, both contributions are zero. For $d \geq b$, Lemma 4 gives

$$\begin{bmatrix} d \\ b \end{bmatrix}_q - q^{d-b} \begin{bmatrix} d-1 \\ b-1 \end{bmatrix}_q = \begin{bmatrix} d-1 \\ b \end{bmatrix}_q.$$

Therefore

$$C_b(q) = \sum_{w \in S_n} q^{M-bd+\binom{b}{2}} \begin{bmatrix} d-1 \\ b \end{bmatrix}_q.$$

The Gaussian binomial $\begin{bmatrix} d-1 \\ b \end{bmatrix}_q$ is zero unless $d-1 \geq b$, i.e. unless $d \geq b+1$. Hence

$$C_b(q) = \sum_{\substack{w \in S_n \\ \text{des}(w) \geq b+1}} q^{\text{maj}(w)-b \text{des}(w)+\binom{b}{2}} \begin{bmatrix} \text{des}(w)-1 \\ b \end{bmatrix}_q,$$

which completes the induction. □

3.8 Proof of the headline theorem

Proof of Theorem 1. The equality $c_{\emptyset}(n) = 1$ is part of Proposition 6.

Fix $b \geq 0$. Proposition 6 gives

$$C_b(q) = \sum_{\substack{w \in S_n \\ d=\text{des}(w) \geq b+1}} q^{\text{maj}(w)-bd+\binom{b}{2}} \begin{bmatrix} d-1 \\ b \end{bmatrix}_q.$$

The switching back-and-forth between $\text{maj}(w)$ and M and $\text{des}(w)$ and d is distracting, but especially in a case like the above where one abbreviation is used and one isn't.

By the definition of the Gaussian binomial coefficient,

$$\begin{bmatrix} d-1 \\ b \end{bmatrix}_q = \sum_{\rho \subseteq b \times (d-b-1)} q^{|\rho|},$$

because $(d-1) - b = d - b - 1$. Therefore

Of all of the steps in this paper to clarify, $(d-1) - b = d - b - 1$ is a funny one.

$$C_b(q) = \sum_{\substack{w \in S_n \\ d = \text{des}(w) \geq b+1}} \sum_{\rho \subseteq b \times (d-b-1)} q^{\text{maj}(w) - bd + \binom{b}{2} + |\rho|}.$$

On the other hand,

$$C_b(q) = \sum_{a \geq 1} c_{(a, 1^b)}(n) q^a.$$

Thus the coefficient of q^a in $C_b(q)$ is exactly the number of pairs (w, ρ) such that

$$w \in S_n, \quad d = \text{des}(w) \geq b+1,$$

$$\rho \subseteq b \times (d-b-1),$$

and

$$a = \text{maj}(w) - bd + \binom{b}{2} + |\rho|.$$

This is precisely the boxed formula in the theorem.

It remains only to verify the stated edge cases. First, the exponent just displayed is always positive under the condition $d \geq b+1$. Indeed, if the descent set of w has size d , say

$$\text{Des}(w) = \{i_1 < \dots < i_d\},$$

then $i_j \geq j$ for every j , so

$$\text{maj}(w) = i_1 + \dots + i_d \geq 1 + \dots + d = \binom{d+1}{2}.$$

Consequently,

$$\text{maj}(w) - bd + \binom{b}{2} \geq \binom{d+1}{2} - bd + \binom{b}{2} = \binom{d-b+1}{2}.$$

Since $d \geq b+1$, the last binomial coefficient is at least 1, and adding $|\rho| \geq 0$ keeps the exponent positive.

When $b = 0$, the rectangle $0 \times (d - 1)$ contains only the empty partition. The condition $d \geq b + 1$ becomes $\text{des}(w) \geq 1$, and the equation for a becomes $a = \text{maj}(w)$. Hence

$$c_{(a)}(n) = \#\{w \in S_n : \text{des}(w) \geq 1, \text{maj}(w) = a\} \quad (a \geq 1).$$

The identity permutation has $\text{des} = 0$ and $\text{maj} = 0$, and its contribution is the separate term $c_{\emptyset}(n) = 1$.

Finally, every permutation in S_n has at most $n - 1$ descents. If $b \geq n - 1$, then the required inequality $d = \text{des}(w) \geq b + 1$ is impossible. Therefore

$$c_{(a,1^b)}(n) = 0 \quad \text{for all } b \geq n - 1.$$

The theorem is proved. □

References

- [1] W. Fulton, *Young Tableaux: With Applications to Representation Theory and Geometry*, London Mathematical Society Student Texts, vol. 35, Cambridge University Press, 1997.
- [2] J. Haglund, B. Rhoades, and M. Shimozono, *Hall–Littlewood expansions of Schur delta operators at $t = 0$* , Séminaire Lotharingien de Combinatoire **79** (2019), Article B79c.
- [3] J. Lentfer, *Diagonal Supersymmetry for Coinvariant Rings*, arXiv:2505.14885, 2025. <https://arxiv.org/abs/2505.14885>
- [4] S. Murai, B. Rhoades, and A. Wilson, *A proof of the Fields Conjectures*, arXiv:2505.24027, 2025. <https://arxiv.org/abs/2505.24027>